Turn Zero: The Noise of Convention When Experts Disagree

Lucas Brennan-Almaraz [1]

Abstract

In competitive Pokémon VGC, each player privately selects which 4 of 6 Pokémon to bring and which 2 to lead—a 90-way decision made from Open Team Sheets before any battle begins. There is no single correct answer: experts facing the same matchup routinely choose differently, and within best-of-three sets the same player changes leads 59% of the time. This makes uncertainty quantification the central challenge—a trainer preparing for a tournament needs to know which leads are viable and when the matchup is too novel for patterns to apply, not just a single prediction. I train a 5-member deep ensemble of permutation-invariant transformers on 382K expert replays. The ensemble covers 50% of expert actions by its 17th-ranked prediction (vs. 45th random), is well-calibrated (adaptive ECE $= 0.012$), and automatically doubles its abstention rate on unseen team archetypes (AUROC $= 0.80$). When all five members agree (5.8% of examples), accuracy reaches 16.6%—nearly $3\times$ the aggregate; when they fully disagree, it drops to 3.1%, indicating no historical consensus. Per-team analysis reveals that predictability is driven by mechanical constraints (commander teams: 50% top-1) rather than strategic reasoning (flexible teams: 0% top-1). Opponent ablation shows matchup conditioning contributes only 1.3 percentage points—the model primarily learns team-specific convention—and conditioning on best-of-three game history recovers +3.6pp on later games, primarily after wins, confirming an asymmetric adaptation ceiling.

CS229: Machine Learning — Stanford University, Winter 2025–26. [1]Department of Computer Science, Stanford University, Stanford, CA, USA. Correspondence to: Lucas Brennan-Almaraz <leba@stanford.edu>.

1. Introduction

Pokémon Video Game Championships (VGC) is the official doubles format for competitive Pokémon. Under Open Team Sheet (OTS) rules, both players see each other's full roster—species, moves, items, abilities—before the battle. Each player then privately selects 4 of 6 to bring and 2 of those 4 to lead: $\binom{6}{4} \times \binom{4}{2} = 90$ legal actions.

The interesting property of this decision is its *multi-modality*. Top players facing the same matchup regularly choose differently, and a reasonable argument exists for each choice. Accuracy still matters—a model that learns real matchup signal is better than one that guesses—but in a domain where experts genuinely disagree, a calibrated *distribution* over actions is more useful than any single prediction. A player preparing for a tournament does not need the "correct" lead; they need to know which leads are viable, how confident the evidence is, and when the matchup is too novel for historical patterns to apply. This makes uncertainty quantification the higher-value challenge.

I present TurnZero, which treats team preview as 90-way classification with an uncertainty quantification stack built on deep ensembles. The system is built on four claims:

1. Per-team analysis reveals that the 6.4% aggregate top-1 masks a heterogeneous landscape: mechanically constrained teams reach 50% top-1 while flexible teams reach 0%, driven by game-rule constraints (Commander ability), not strategic reasoning (speed control: $\Delta H = 0.016$ nats, null).

2. Within-set adaptation creates irreducible label noise—players change leads 59% of the time between games, 72% after a loss—and a +34% parameter architecture ablation confirms the model is at the noise ceiling, not the capacity ceiling. Opponent ablation confirms only 1.3pp comes from matchup conditioning; conditioning on best-of-three game history recovers +3.6pp on later games primarily after wins, establishing an asymmetric adaptation ceiling at $\sim$9.4% top-1.

3. A permutation-invariant transformer ensemble trained on 382K replays learns matchup-specific signal above baselines: its top-17 predictions cover 50% of expert

actions (2.6× uniform), and distributional evaluation (JSD) confirms it is a genuine mixed-strategy estimator that beats all baselines including team-specific marginals. The architecture avoids by construction the positional leakage that inflated prior work from 99.9% to 79% after correction.

4. Standard UQ techniques (calibration, selective prediction, OOD detection via mutual information) are applied; the ensemble is well-calibrated (adaptive ECE $= 0.012$) though typical confidence is low (median 4.5%) and calibration degrades on OOD data (ECE: $0.012 \rightarrow 0.076$).

No prior work isolates VGC team preview as a standalone prediction problem with uncertainty quantification.

2. Related Work

Full-battle agents. VGC-Bench (Angliss et al., 2026) defines the same 90-way action space and trains transformer policies via behavior cloning on 705K battle logs, but team preview is one incidental step in a full-game RL pipeline—never isolated or evaluated, and no UQ is reported. EliteFurretAI (Simpson, 2024) initially reported 99.9% team preview accuracy, later traced to positional memorization: Showdown logs store Pokémon in a fixed order, and 88.6% of unique states mapped to exactly one action. After random-order augmentation, accuracy dropped to 79%. My architecture avoids this by design via canonical sorting and mean pooling. Metamon (Grigsby et al., 2025) and PokeChamp (Karten et al., 2025) target Singles—a different format with no team preview decision.

Lead prediction. Carli (Carli, 2025) is the only prior work to specifically predict VGC leads, using SVD-based similarity matching on ∼5K logs with species-only features and no formal evaluation metrics or UQ. TurnZero differs in every dimension: supervised 90-way classification with full OTS features, 382K examples, and uncertainty quantification.

Set architectures. The permutation-invariant design draws on Deep Sets (Zaheer et al., 2017) and Set Transformer (Lee et al., 2019), using self-attention over unordered token sets with mean-pool aggregation.

3. Data and Problem Setup

Action space. Each player chooses 4 of 6 Pokémon to bring ($\binom{6}{4} = 15$) and arranges 2 of those 4 as leads ($\binom{4}{2} = 6$), yielding $15 \times 6 = 90$ distinct actions encoded as integers via a canonical bijection.

Dataset. 212K best-of-three Regulation G tournament battles from Pokémon Showdown (Angliss et al., 2026) yield 382K directed examples after deduplication. Each example pairs two full Open Team Sheets—species, item, ability, tera type, and all four moves for each of 12 Pokémon—with a ground-truth action.

Label observability. Leads are always visible in the log. The full bring-4 is only observable when all 4 Pokémon appear during the game (∼80% of examples, denoted Tier 1). All test metrics use Tier 1 data only.

Split design. Teams are clustered by species composition (≥4-of-6 overlap → connected components via union-find, 7,826 clusters). *Regime A* (in-distribution) holds out Team A variants within clusters: 247K train / 35K val / 40K test, all 90 actions in every split. *Regime B* (OOD) holds out entire clusters to test distribution-shift sensitivity. Identical (team_a, team_b, action) triples are deduplicated across splits, but the same matchup with different expert actions is retained—this is signal (multi-modality), not leakage.

4. Methods

4.1. Baselines

Popularity (0 params): predict the global action frequency distribution from training data. This captures metagame convention without matchup reasoning.

Logistic regression (∼4K params): bag-of-embeddings over 12 Pokémon's discrete features through a linear layer to 90-way softmax. The logistic baseline achieves higher top-1 than popularity (4.0% vs. 1.3%) but *worse* NLL (4.580 vs. 4.497)—it learns some signal but assigns overconfident probabilities to wrong actions, illustrating that accuracy and calibration can diverge.

4.2. Transformer Ensemble

Each of the 12 Pokémon is tokenized as the sum of 8 learned embeddings (species, item, ability, tera type, 4 moves) and processed by a 4-layer transformer encoder (d=128, H=4, d_{ff}=512, dropout 0.1; 1.16M parameters). The architecture is permutation-invariant: tokens are canonically sorted and aggregated via mean pooling before a linear head. No positional encoding is used—order carries no information and would introduce the leakage risk documented in (Simpson, 2024).

I train 5 independent members with different random seeds (Lakshminarayanan et al., 2017). The ensemble averages probabilities: $\bar{p}(a \mid x) = \frac{1}{5}\sum_m p_m(a \mid x)$. Total uncertainty decomposes into predictive entropy $\mathcal{H}[\bar{p}]$ (data noise plus model ignorance) and mutual information $\mathrm{MI} = \mathcal{H}[\bar{p}] - \frac{1}{5}\sum_m \mathcal{H}[p_m]$ (epistemic uncertainty from member disagreement). Training uses cross-entropy, AdamW (lr=3×10^{-4}), cosine annealing, and BF16 mixed precision. Each member converges in ∼8 epochs (∼5 min on a single RTX 4080 Super).

Table 1. Test performance (Regime A, Tier 1). Top-k accuracy (%, ↑), NLL (↓), adaptive ECE (↓, equal-mass binning (Nixon et al., 2019)). Bootstrap 95% CIs are cluster-aware (B=1000).

Model	Params	Top-1	Top-3	NLL	ECE
Random	—	1.1	3.3	4.500	—
Popularity	0	1.3	3.9	4.497	.001
Logistic	∼4K	4.0	10.1	4.580	.059
Transformer	1.16M	5.5	14.0	4.105	.016
Ensemble	**5.8M**	**6.4**	**15.5**	**4.031**	**.012**
95% CI		[2.6, 6.6]	[6.8, 16.0]	[4.01, 4.43]	[.002, .013]

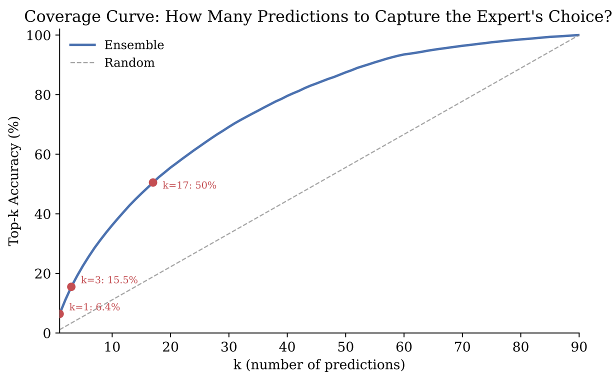

Figure 1. Cumulative coverage vs. k. The ensemble covers 50% of expert actions by k=17 (vs. k=45 random), narrowing the space from 90 to 17 actions for half-coverage.

4.3. Selective Prediction

Given a confidence threshold τ, the system abstains when $\max_a \bar{p}(a) < \tau$, trading coverage for accuracy (Geifman and El-Yaniv, 2017).

5. Results

All results are on Regime A test data, Tier 1 only (n=32,328), unless noted.

5.1. Aggregate Performance

Table 1 shows that the ensemble achieves 6.4% top-1 and 15.5% top-3 on the 90-way task (5.8× and 4.7× random). The NLL improvement over popularity (4.031 vs. 4.497, where $\ln 90 \approx 4.50$ is the maximum-entropy baseline) confirms the model learns matchup-specific signal beyond metagame frequencies, though the absolute improvement is modest on this scale. The wide bootstrap CIs (top-1: [2.6%, 6.6%]) are not a weakness of the model but a property of the task—performance varies dramatically across team archetypes (Section 5.5).

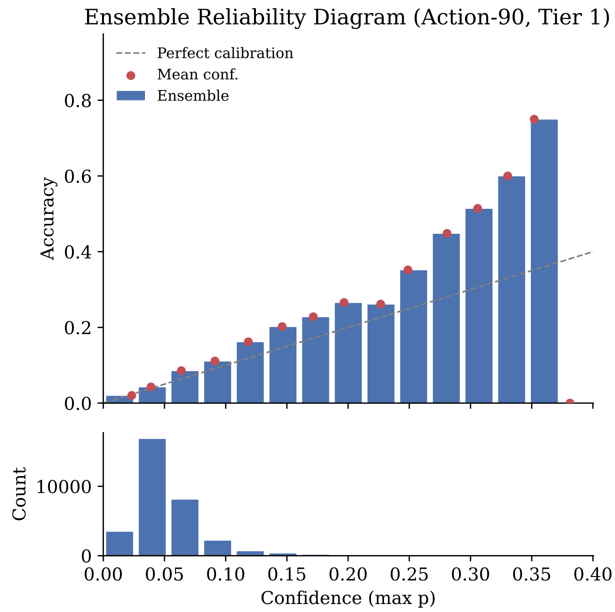

Figure 2. Predicted confidence tracks observed accuracy (adaptive ECE = 0.012). A caveat: only 6% of predictions exceed 10% confidence, so the calibrated output is typically a gently peaked distribution rather than a sharp recommendation.

5.2. Beyond Top-1: Coverage in a Multi-Modal Space

Top-1 accuracy is a limited metric for a multi-modal task. Figure 1 shows that the ensemble needs only 17 predictions to cover 50% of expert actions—2.6× better than uniform. By k=10, coverage already reaches 36%.

The marginal decomposition reveals where difficulty concentrates. Bring-4 and lead-2 marginals each achieve ∼18% top-1 and ∼43% top-3 over their respective 15 classes. But conditional on a correct bring-4, the model picks the right lead pair only 32% of the time (2× the 16.7% random baseline for a 6-way choice). Experts agree on *what* to bring but disagree about *who leads*—bringing is constrained by role coverage (you need answers to specific threats), while leading is a tempo and read decision with more degrees of freedom.

5.3. Calibration

Figure 2 shows the ensemble's predicted confidence closely tracks observed accuracy (adaptive ECE = 0.012 with equal-mass binning (Nixon et al., 2019)). Post-hoc temperature scaling yielded $T = 1.158$ (near-identity), confirming the ensemble is already well-calibrated; it is omitted from the final system. A caveat: on a 90-class problem with high intrinsic uncertainty, most predictions are near-uniform (median confidence 4.5%, only 6% exceed 10%), so calibration is most informative for the minority of predictions where

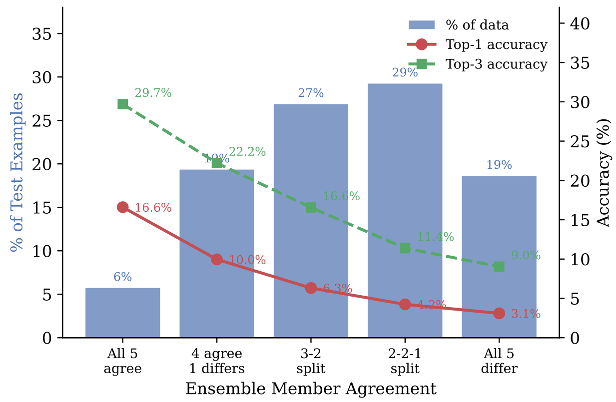

Figure 3. When all 5 ensemble members agree on the same top-1 prediction (5.8% of examples), accuracy reaches 16.6% top-1 / 29.7% top-3. When all 5 disagree, accuracy drops to 3.1% / 9.0%. Member agreement is a direct, interpretable confidence signal.

the model concentrates mass on a few actions.

5.4. Ensemble Agreement and Distribution Shift

Figure 3 shows that ensemble agreement correlates monotonically with accuracy. When all 5 members agree on the same prediction (5.8% of test examples), top-1 accuracy reaches 16.6%—nearly 3$\times$ the aggregate and 15$\times$ random on a 90-way task. When all 5 disagree (18.7%), accuracy drops to 3.1%, near random. The monotonic relationship makes member agreement a useful confidence signal, and the uncertainty decomposition into predictive entropy vs. mutual information provides a basis for selective abstention.

Distribution-shift sensitivity. On Regime B (held-out clusters), abstention jumps from 20% to 46%. The ensemble automatically becomes more cautious on unfamiliar teams, without any explicit OOD mechanism—mutual information from member disagreement naturally separates in-distribution from novel compositions (AUROC $= 0.80$). Counterintuitively, accuracy on non-abstained Regime B examples is *higher* (11.7% vs. 6.4%)—a selection effect: OOD teams that survive the confidence filter are effectively "self-solving," teams whose mechanical constraints are strong enough that the model infers the correct lead from structure alone, without ever having seen a training example. Calibration degrades on OOD data (ECE: $0.012 \rightarrow 0.076$), a genuine limitation partially compensated by the higher abstention rate.

5.5. Per-Team Predictability

The 6.4% aggregate top-1 is a misleading average. Figure 4 groups the test set by species composition (153 archetypes with $\geq$20 examples) and reveals a strong correlation between ensemble entropy and accuracy ($r = -0.56$ for top-3). The correlation strengthens when restricted to well-sampled archetypes ($\rho = -0.77$ for $n \geq 50$, 51 teams), confirming it is not a small-sample artifact.

The extremes are domain-plausible. The most predictable teams are "commander" archetypes (e.g., Dondozo + Tatsugiri, $H = 3.10$, 50% top-1) where the Commander ability mechanically requires both Pokémon to lead together—the game rules concentrate the action distribution into a single dominant play. The least predictable are flexible "goodstuffs" teams ($H = 4.41$, 0% top-1) with many viable configurations. Per-team accuracy tracks action-mode frequency ($r = 0.55$): the model learns *which teams have solved lead selection*, not opponent-conditional strategy.

Speed control mode (Trick Room vs. Tailwind) does not explain this—the entropy gap is 0.016 nats, effectively zero. Predictability is driven by mechanical constraints, not strategic choices.

Table 2. Feature masking stress test. Each row masks a cumulative set of features and re-evaluates the frozen ensemble.

Masked features	**Top-1**	**Top-3**	**NLL**
None (baseline)	6.4	15.5	4.031
Items	4.5	12.2	4.188
Tera type	5.1	12.2	4.177
2 of 4 moves	3.1	8.1	4.388
All 4 moves	1.0	3.2	4.714
All except species	1.2	3.4	4.672

5.6. Feature Importance

Table 2 shows that moves carry the dominant signal. Masking all four drops top-3 from 15.5% to 3.2%—near the popularity baseline's 3.9%. Items and tera type each cost $\sim$3pp individually. The "all except species" row shows that species names alone carry almost no predictive signal beyond random (1.2% vs. 1.1%)—unsurprising because species is merely a label; the game-relevant information is fully captured by moves, ability, item, and tera type.

The mirror/non-mirror gap reinforces this picture: top-1 is 6.8% on mirror matchups (common archetypes with established lead patterns) but 3.8% on novel ones—the model learns convention, and convention requires repeated observation.

5.7. Opponent Ablation

Figure 5 shows that removing the opponent's team at inference time costs only 1.3pp top-1 ($6.4\% \rightarrow 5.1\%$, NLL $4.031 \rightarrow 4.125$), while removing the player's own team collapses accuracy to 1.1% (NLL 4.730)—indistinguishable from uniform random. The model retains $\sim$80% of its performance from self-team features alone, learning *what this*

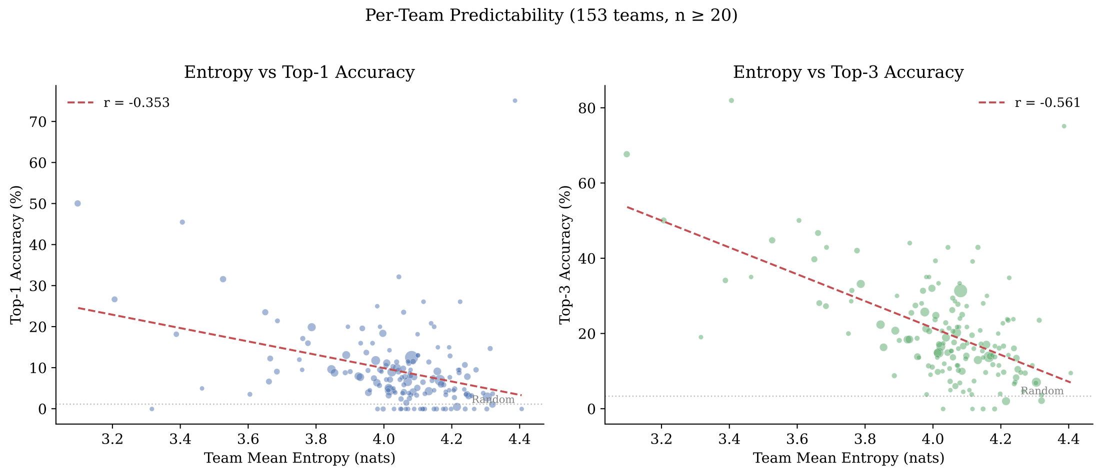

Figure 4. Mean ensemble entropy vs. top-3 accuracy for 153 team archetypes ($\geq$20 Tier 1 examples). Point size $\propto n$. $r = -0.56$: teams with concentrated action distributions are easy to predict; those spreading across many viable plans are not.

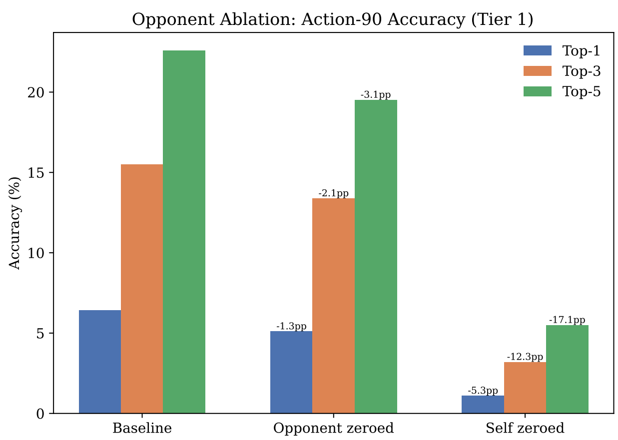

Figure 5. Removing the opponent's team (all features $\rightarrow$ UNK) costs only 1.3pp top-1; removing the player's own team collapses accuracy to random. Essentially all predictive signal resides in the player's own team composition.

team usually does rather than *how it responds to a specific opponent*. This is the strongest direct evidence that the accuracy ceiling is structural: even a perfect matchup-conditional model would gain little, because the opponent signal contributes only $\sim$1.3pp above marginal team habits.

5.8. Distributional Evaluation

Top-1 accuracy penalizes a model for distributional correctness in a multi-modal domain. Table 3 evaluates each method as a *distributional* estimator via Jensen-Shannon divergence against the empirical population distribution on well-sampled matchups. The ensemble beats all baselines at both granularities; the gap over the team-A marginal (0.328 vs. 0.359 on action-90) confirms the model learns some matchup-conditional structure, even though opponent ablation shows this contributes little to point accuracy. The model extracts a small but real distributional signal from matchup context that does not manifest as top-1 improvement because the action space is too entropic for distributional gains to concentrate on the mode.

Table 3. Jensen-Shannon divergence (JSD, $\downarrow$) between each method's predicted distribution and the empirical population distribution, computed on well-sampled matchups ($n \geq 10$).

Method	Action-90	Lead-2
Ensemble	**0.328**	**0.159**
Team-A marginal	0.359	0.183
Popularity prior	0.480	0.261
Uniform	0.484	0.264

6. Discussion

The ceiling is multi-modality, not model capacity. Across 53K linkable best-of-three sets (136K game-to-game transitions), players change their lead pair 59% of the time but their bring-4 roster only 52%—and lead changes rise to 72% after a loss versus 46% after a win ($\chi^2 = 10{,}126$, $p < 10^{-6}$). This confirms the marginal decomposition from a behavioral angle: even the same player treats roster composition as more stable than lead arrangement. The training set contains genuine label contradictions: the same matchup produces different leads not because experts disagree, but because the *same* player adapts to information revealed in prior games. No turn-zero model can capture this within-set adaptation, establishing a principled floor on irreducible label noise. Replacing the flat self-attention architecture with a hierarchical cross-attention variant (1.56M params, +34%)

produced identical accuracy ($\pm$0.3pp), consistent with this ceiling. Paradoxically, stronger players likely make the problem *harder*: a player who always leads the same comfort pick is trivially predictable, while one who adapts to reads, metagame shifts, and mid-set information introduces exactly the kind of decision variance that no team-sheet-only model can resolve.

Convention, not strategy—but the distribution is genuine. The mirror/non-mirror gap (6.8% vs. 3.8%), the entropy-accuracy correlation, and the opponent ablation (-1.3pp) all point to the same conclusion: the model learns what experts *typically* do with familiar teams, not what they *should* do against specific opponents. Stratifying by population consensus sharpens this: on high-consensus matchups the ensemble reaches 14.8% top-1 and 37.0% top-3; on zero-consensus matchups it drops to 0%. The model is not failing—it correctly reflects population-level uncertainty. This reframes "6.4% is bad" into "6.4% is what distributional accuracy looks like when experts genuinely disagree." Distributionally, the ensemble beats the team-A marginal (JSD 0.328 vs. 0.359) even though point accuracy barely changes when the opponent is removed—the model extracts real matchup-conditional signal that improves distributional fit without concentrating enough mass to move the mode.

Engineering matters. The leakage-safe architecture (avoiding the 99.9%$\rightarrow$79% collapse in (Simpson, 2024)), Tier 1/2 observability delineation, and cluster-aware bootstrap had outsized impact on producing trustworthy results. These methodological choices, not architecture, were the hard problems.

Within-set history partially breaks the ceiling. A natural question is whether conditioning on prior game context—outcome and observed leads—can recover some of the "irreducible" label noise. A sequential variant (+2,816 parameters, $<$0.3% overhead) that embeds game number, prior result, and prior lead pair as a context token recovers +3.55pp on Games 2–3 (5.87% $\rightarrow$ 9.42% top-1; Figure 6), but with a striking asymmetry: after wins the gain is +6–8pp (winners repeat what worked), while after losses it is $\leq$1pp (+1.1pp on Game 2, -0.1pp on Game 3)—losers improvise unpredictably. Game 1 accuracy is unchanged (6.41% vs. 6.64%), confirming the sentinel mechanism introduces no phantom signal. Even with history, the sequential model reaches only 9.4% top-1—the ceiling is lowered but not broken. The after-loss subset, where 72% of lead changes occur, remains genuinely irreducible: the most adaptation-heavy games are where prediction fails, connecting the statistical observation (label noise) to a concrete behavioral mechanism (winners repeat, losers improvise).

Limitations. Bootstrap CIs are wide (top-1: [2.6%, 6.6%]). Calibration degrades on OOD data (ECE: $0.012 \rightarrow 0.076$). The model underperforms a per-team mode lookup by 6.8pp across 153 archetypes; its potential value over simple frequency tables lies in calibrated probabilities across the full 90-action space, but this remains unvalidated in deployment. Training data is exclusively best-of-three tournament play; ladder games were deliberately excluded to match deployment conditions but limit data scale. Results are specific to Regulation G; whether multi-modality patterns generalize to other formats is untested.

Contributions. Solo project. Code: https://github.com/leba01/turnzero

Addendum: Additional Observations

The following findings informed the design but did not fit the main narrative.

A1. Label noise is tolerable. Roughly 20% of training examples have fabricated back-2 labels (the parser fills unobserved slots with a deterministic heuristic). Three loss modes were compared across 5-member ensembles: (a) standard cross-entropy on all data, (b) a multi-task loss that marginalizes Tier 2 labels to lead-2 probabilities via a 90×15 aggregation matrix in probability space, and (c) training on Tier 1 only. Differences were $\leq$0.6pp top-1 (a: 6.4%, b: 6.5%, c: 5.9%). Dropping 20% of data hurts more than the noise does.

A2. The metagame is a small-world network. The species-overlap clustering ($\geq$4-of-6 shared $\rightarrow$ connected components) places 91% of test data in a single mega-cluster. This is not a bug: hub species (Incineroar 37%, Urshifu-Rapid-Strike 38%, Rillaboom 29%) create transitive bridges through the union-find, producing one giant connected component. Only 1.8% of random team pairs share $\geq$4 species directly, but the small-world structure connects nearly everything. This motivated the switch to exact species-6 composition for per-team analysis (153 archetypes with $\geq$20 Tier 1 examples).

A3. Per-team extremes. The most predictable teams are mechanically constrained: Dondozo commander variants reach 50% top-1 ($H = 3.10$), and a Trick Room team with Lunala/Torkoal reaches 32% ($H = 3.53$). At the other extreme, 22 of 153 teams (14%) have 0% top-1 accuracy—the model never predicts the expert's exact action. The worst large team is Gothitelle/Incineroar/Kyogre/Raging Bolt/Rillaboom/Scream Tail (n=193, 0.5% top-1, $H = 4.22$).

A4. Confidence in a 90-class world. Mean confidence (max predicted probability) is 5.3%, median 4.5%, maximum 38.1%. Only 6% of predictions exceed 10% confidence; 0.8% exceed 20%. This is expected for a 90-class problem with high intrinsic uncertainty—even the model's "most confident" prediction assigns less than 40% to a single action.

A5. Sequential model by stratum. Figure 6 breaks down the base vs. sequential model comparison by game number and prior outcome. The asymmetry is stark: after wins the sequential model nearly doubles top-1 accuracy, while after losses it provides little benefit.

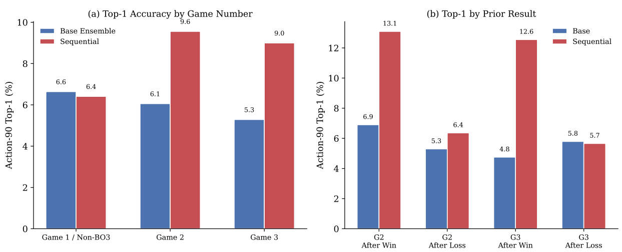

Figure 6. Base vs. sequential model top-1 accuracy by game number and prior outcome. The +3.55pp gain on Games 2–3 comes almost entirely from post-win games; post-loss games remain near baseline.

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